USB POWER S9.1 / SINGLE BOARD, +/−12V ONLY, 15 V PD INPUT

Own−input enable: VIN pulse clamp; split U2 grounds; STARTED one−shot; no output crowbar.

USB PD, main eFuse and VIN damper



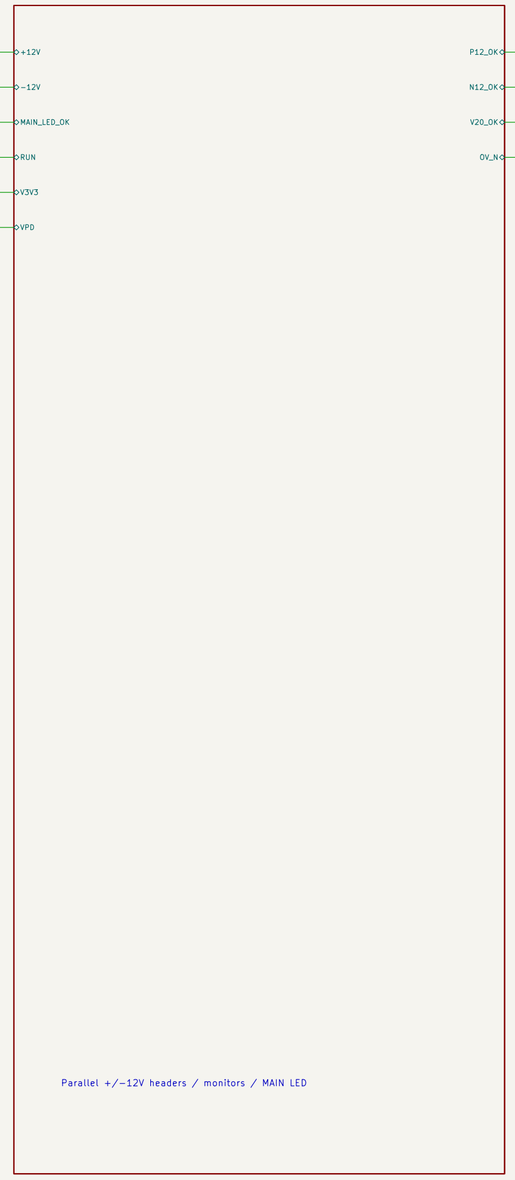
File: 01_usb_input.kicad_sch

+12 V buck−boost converter



File: 02_positive_12v.kicad_sch

Two output headers − monitors, overvoltage guard and MAIN LED



File: 05_outputs_status.kicad_sch

−12 V inverting converter



File: 03_negative_converter.kicad_sch

ONE−SHOT FAULT POLICY
STARTED retains the first start within each power epoch.
Faults cannot restart it: full power reset CAN. Repeated source resets FAIL qualification.
OV_N presets FAULT asynchronously. Crowbar vs eFuse deferred/open.

+12V 2.4 A / −12V 0.8 A, <=33 W together (prototype, not measured).
13.5−16.5 V at module: 15 V / 3 A PD. Rack +12V <=0.44 mF. Impedance bounds in calcs.
No reverse−current blocking; no powered parallel supply or unrestricted hotplug qualification.

Always−on control − fault latch, overvoltage preset and RESET



File: 04_fault_reset.kicad_sch

One−shot arming latch (STARTED): U18/U20 on VPD: see docs/s8−single−board.md
S9.0 single board: +/−12V only (+5V subsystem removed); MAIN LED only

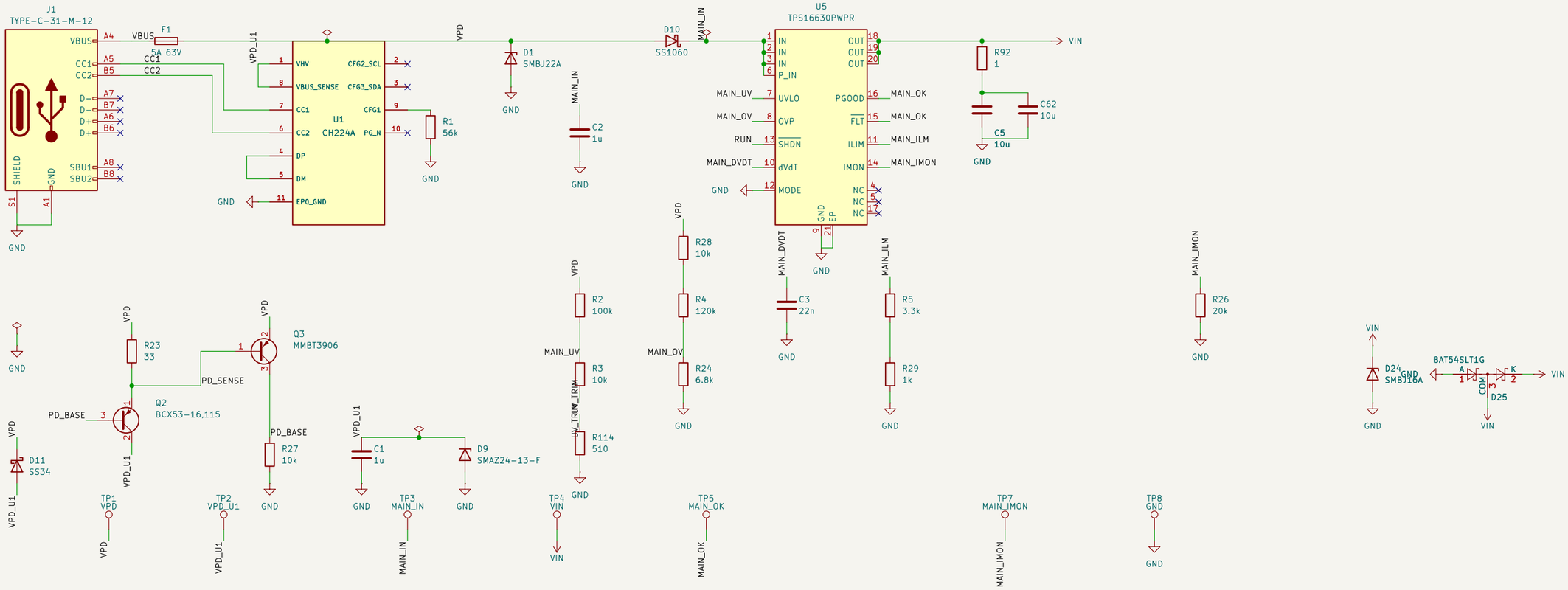
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Sheet: /
File: usb−power−45w−single.kicad_sch

Title: USB power S9 single board − wired power flow overview

Size: User	Date:	**Rev: S9.1**
KiCad E.D.A. 10.0.6		Id: 1/6

USB PD, MAIN EFUSE AND VIN DAMPER



S9.1: 15 V / 3 A PD. R1 56k. U5 UVLO: R2 100k 0.1% / (R3 10k 0.1% + R114 510R 1%).
Worst start <13.0 V; range 13.5–16.5 V at module incl. <=0.75 V cable drop. OVP unchanged.
D24 SMBJ16A VIN pulse clamp (600 W 10/1000 us, 26 V rated clamp): not a continuous shunt.
D25: low-current negative clamp, slew-limited room-temperature test only. See calcs residuals.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇—— MAIN_OK

RUN ◇—— RUN

VIN ◇—— VIN

VPD ◇—— VPD

S9.1 review fixes; P2.1 UNROUTED; docs/s8-single-board.md

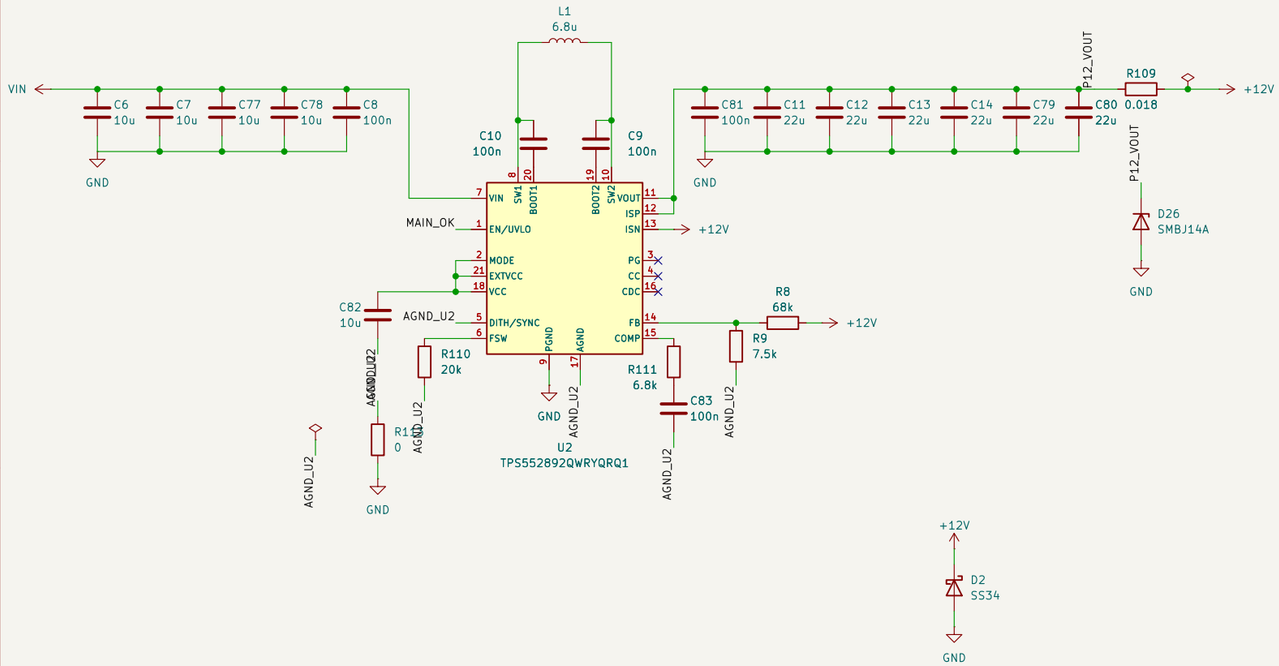
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Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S9.1**
KiCad E.D.A. 10.0.6		Id: 2/6

+12 V BUCK–BOOST CONVERTER



S9.1: TPS552892-Q1, 966 kHz, L1 6.8u; R109 18mR 1W (2.62–2.94 A). Rating +12V 2.4 A, total <=33 W.
AGND_U2: U2.5/.17, R9, R110, C83, C82. ONLY join to PGND (GND) through R113 at C82 ground terminal.
COMP R111 6.8k / C83 100n (zero 234 Hz). Rack <=0.44 mF with bounded cable impedance; see full loop sweep.
R109 is a slow OUTPUT current limit, not a switch limit. Thermal / Bode measurements remain release gates.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇ +12V

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

S9.1 review fixes; P2.1 UNROUTED; docs/s8-single-board.md

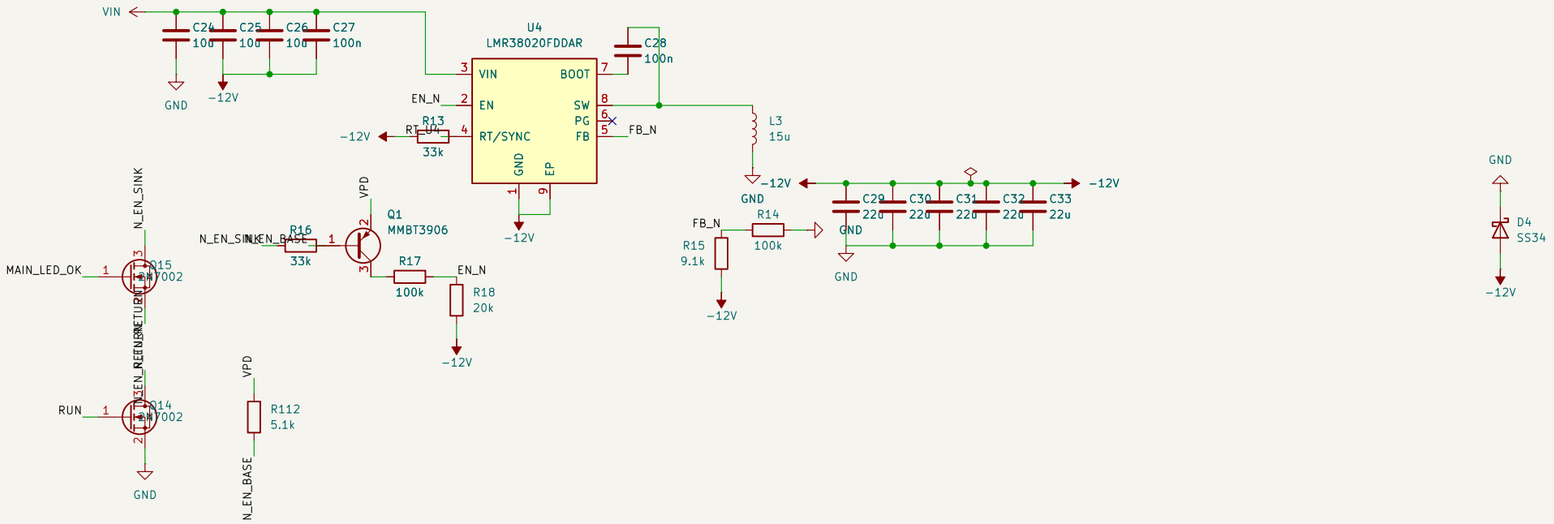
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Sheet: /+12 V buck–boost converter/
File: 02_positive_12v.kicad_sch

Title: +12 V buck–boost converter

Size: User	Date:	**Rev: S9.1**
KiCad E.D.A. 10.0.6		Id: 3/6

−12 V INVERTING CONVERTER



S9.1: U4 enable is powered from own VPD upstream of D10, NEVER from +12V or pumped VIN.
Q14 RUN and Q15 MAIN_LED_OK form an AND sink for Q1 base; R112 returns base to emitter when either is off.
EN_N relative to −12V = (VPD − VCEsat − V(−12V))/6 only when qualified; otherwise R18 holds it at 0 V.
Unpowered or latched-off live +/−12V bus cannot self-enable U4. No reverse-power blocking is claimed.
LMR38020F power stage unchanged. Powered parallel supplies and unrestricted hotplug remain prohibited.

INTER-SHEET INTERFACE (see wired power-flow overview)

−12V −12V

MAIN_LED_OK MAIN_LED_OK

RUN RUN

VIN VIN

VPD VPD

S9.1 review fixes; P2.1 UNROUTED; docs/s8-single-board.md

Big Modular

Sheet: /−12 V inverting converter/
File: 03_negative_converter.kicad_sch

Title: −12 V inverting converter

Size: User	Date:	**Rev: S9.1**
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ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET

U9
TPS7A1633DGNR
C46 1u
C47 10u
R63 10k
MAIN_OK

R64 10k
RUN

U10
TPS3808G01DBVR
R91 15k
R65 47k
R66 10k
POR_SEN
POR_SENSE
RESET_BTN_N
RESET_N
R67 10k
R68 10k
C48 100n

U11
TPS3808G01DBVR
START_SENSE
TIMER_GO
R69 33k
R70 10k
START_CT
C49 100n
R71 10k
ARMED
C50 100n

SW1
TL3340AF160QG
RESET_BTN_N

U12
SN74HC74DR
RESET_N
V3V3
START_N
FAULT_CLK
STARTED
FAULT_LATCH
NOT_FAULT
C52 100n

U13
SN74HC08DR
NOT_FAULT
ARMED
BAD_FILTERED
RUN
STARTED
MAIN_OK
V20_OK
FAULT_CLK
TIMER_GO
MAIN_LED_OK
C53 100n

U14
SN74HC20DR
MAIN_LED_OK
P12_OK
N12_OK
RUN
ANY_BAD
START_N
C54 100n

R72 10k
ANY_BAD
BAD_RC
C51 100n

U15
SN74LVC1G17DBVR
BAD_RC
BAD_FILTERED
C55 100n

TP9 RUN
TP10 ARMED
TP11 FAULT_LATCH
TP12 P12_OK
TP14 N12_OK
TP15 V20_OK
TP16 V3V3
TP6 OV_N
TP19 STARTED

S9.1 retains S8.5 STARTED one-shot: START_N = NAND(MAIN_LED_OK,RUN); TIMER_GO = RUN AND STARTED. U11 -> ARMED after 0.31–0.88 s. RESET clears STARTED/FAULT; U20 OV_N presets FAULT even before ARMED. A full V3V3/POR reset DOES re-arm; qualification rejects repetitive hard-reset sources; no masking by retries. ANY_BAD = NAND(MAIN_LED_OK,P12_OK,N12_OK,V3V3). R83 100k validity floor unchanged.

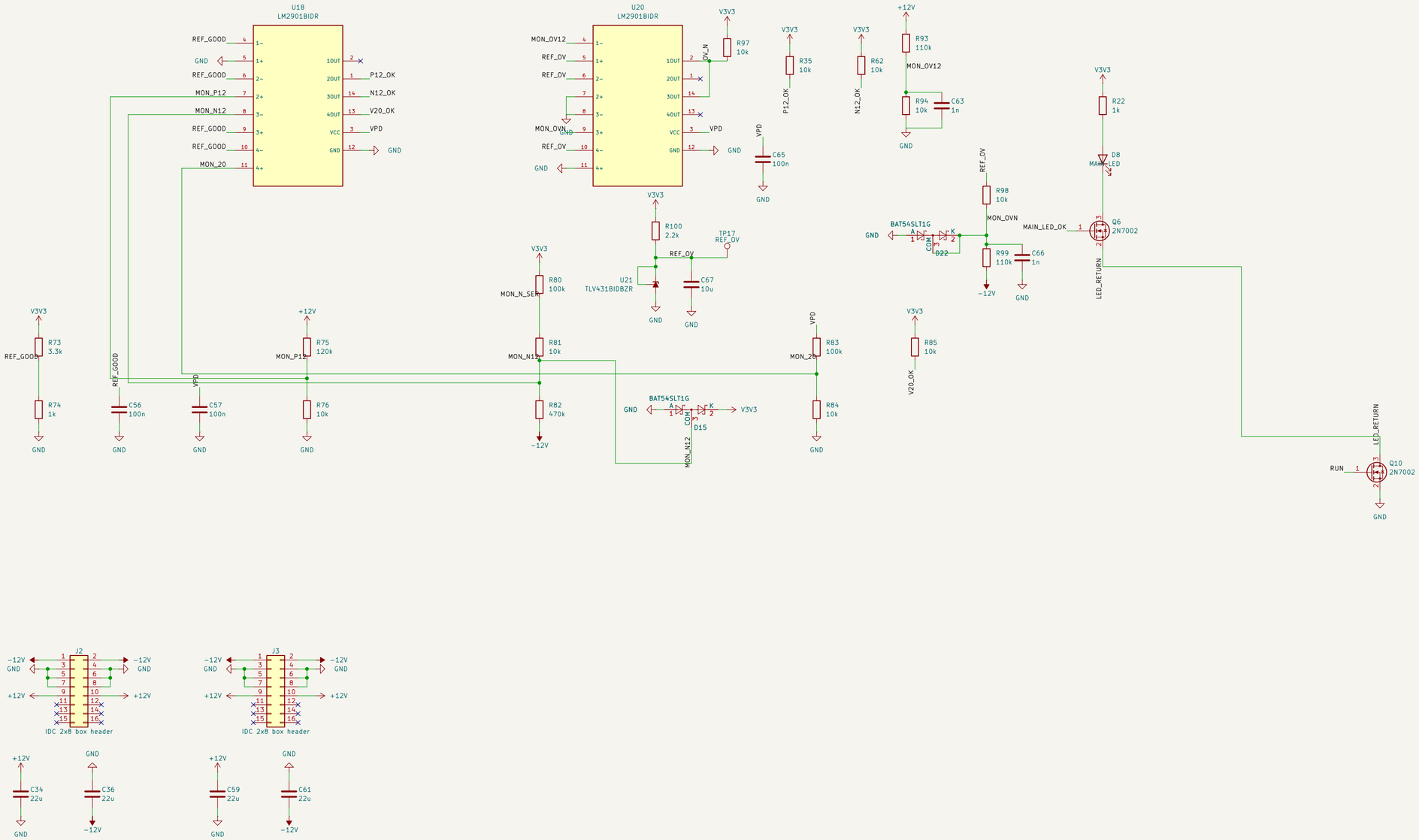
INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_LED_OK
MAIN_OK
N12_OK
OV_N
P12_OK
RUN
V20_OK
V3V3
VPD

S9.1 review fixes; P2.1 UNROUTED; docs/s8-single-board.md
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Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 04_fault_reset.kicad_sch
Title: Always-on control — fault latch, overvoltage preset and RESET
Size: User | Date: | **Rev: S9.1**
KiCad E.D.A. 10.0.6 | Id: 5/6

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

S9.1: +/−12V only. J2/J3 pins 11–16 NC. MAIN LED only. U18/U20 on VPD, own REF_OV unchanged.
Guard +12V 14.48–15.27 V, −12V magnitude 13.27–14.02 V. Guard disconnects INPUT, not an external bus.
No rail crowbar or eFuse. Uninjected rails decay with load; injected rail remains at the external voltage.
See docs/s8-single-board.md for separate latch, clamp, discharge and survival acceptance criteria.



INTER-SHEET INTERFACE (see wired power-flow overview)

+12V — +12V

−12V — −12V

MAIN_LED_OK — MAIN_LED_OK

N12_OK — N12_OK

OV_N — OV_N

P12_OK — P12_OK

RUN — RUN

V20_OK — V20_OK

V3V3 — V3V3

VPD — VPD

S9.1 review fixes: P2.1 UNROUTED; docs/s8-single-board.md

Big Modular

Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 05_outputs_status.kicad_sch

Title: Two output headers — monitors, overvoltage guard and MAIN LED

Size: User	Date:	**Rev: S9.1**
KiCad E.D.A. 10.0.6		Id: 6/6